

IT Security 2

Unit 7 – Creating and managing NAT policy rules

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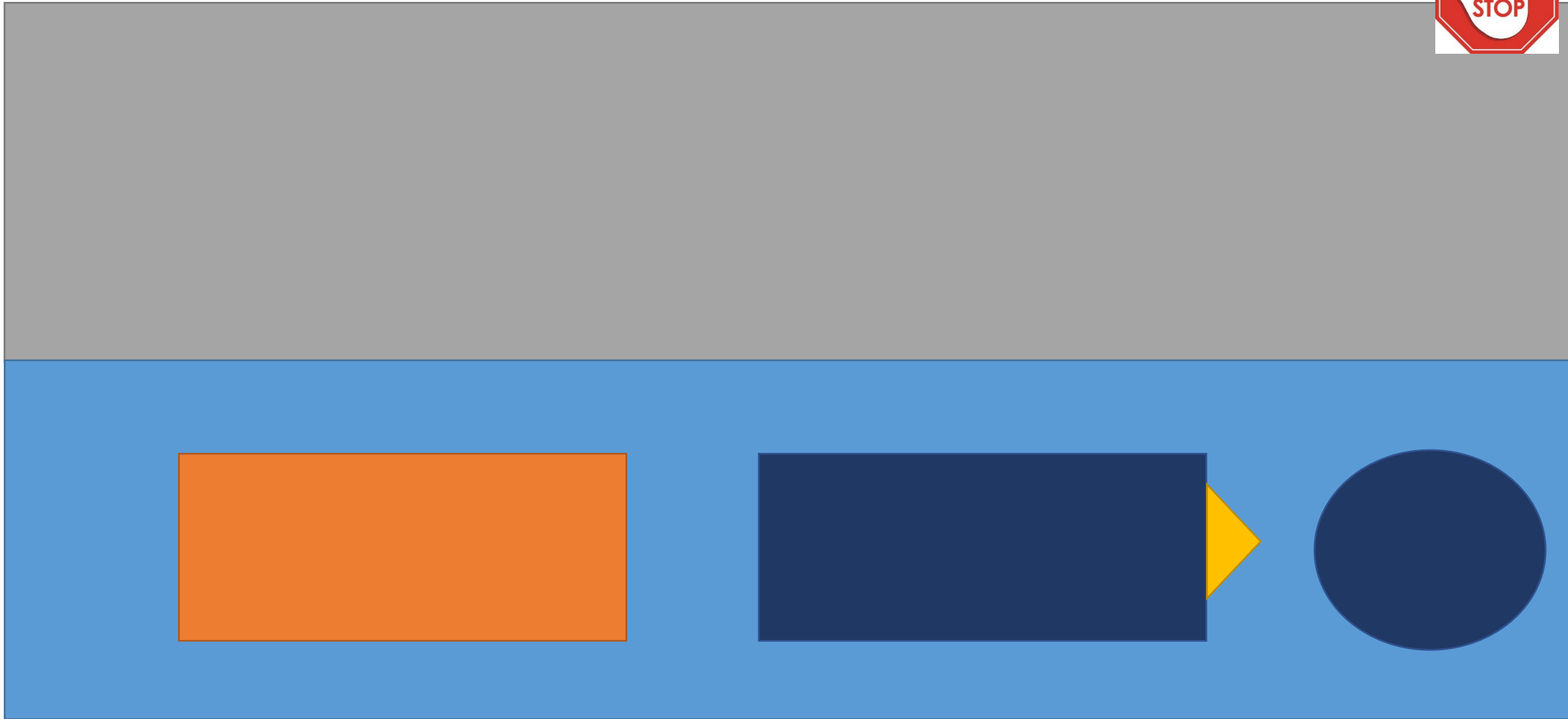
Announcements

- Week 7 – Late Work Policy
- IT Club Event

Creating and managing NAT policy rules

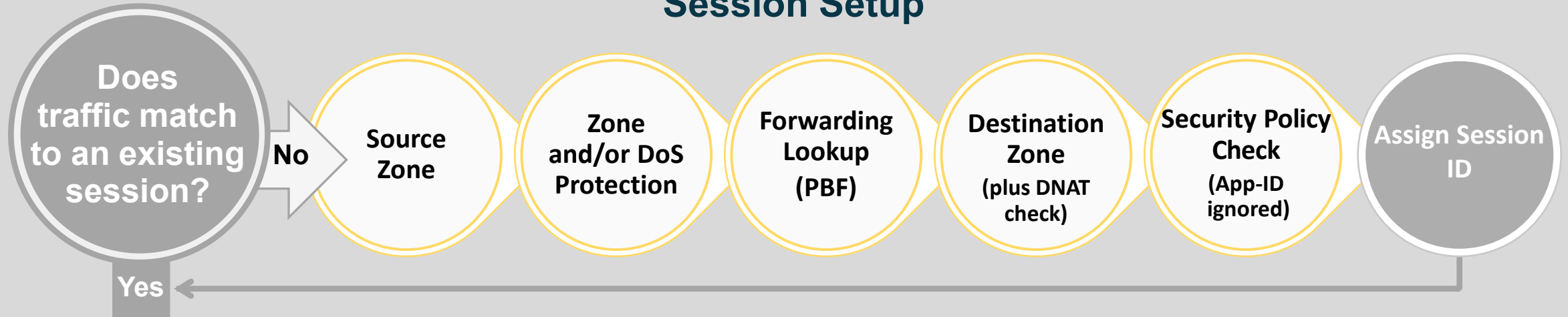
Some slides were adapted from Palo-Alto for use in this class. Content is a mesh between vendor provided information and instructor comments.

Flow Logic of the Next-Generation Firewall

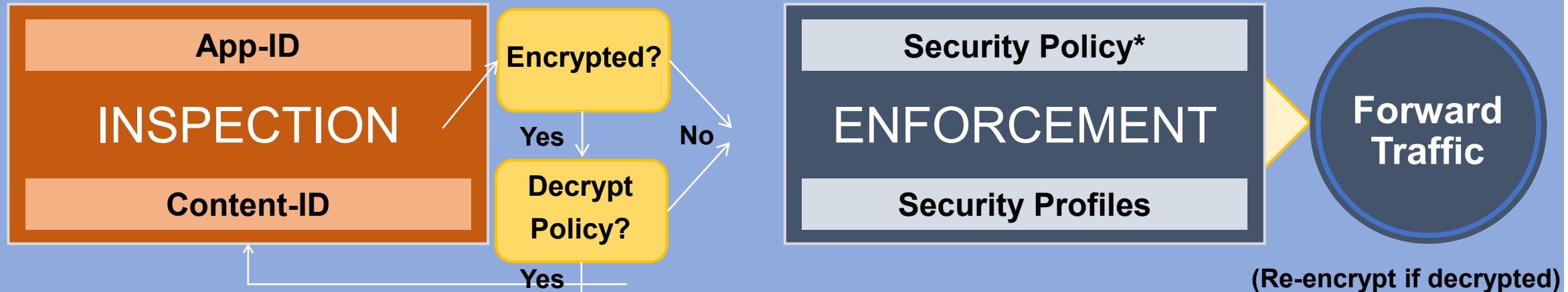


Flow Logic of the Next-Generation Firewall

Session Setup

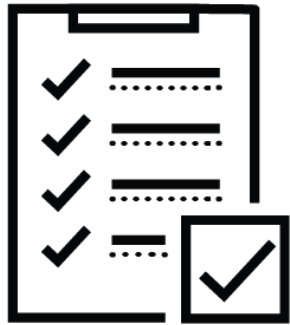


Inspection and Enforcement



* Policy check relies on pre-NAT IP addresses

CREATING AND MANAGING NAT POLICY RULES



GET OUTBOUND TRAFFIC FLOWING

- Network address translation
- Source NAT configuration
- Destination NAT configuration

Learning Objectives

After you complete this module, you should be able to:

- Configure a NAT policy to implement source NAT
- Configure a NAT policy to implement destination NAT





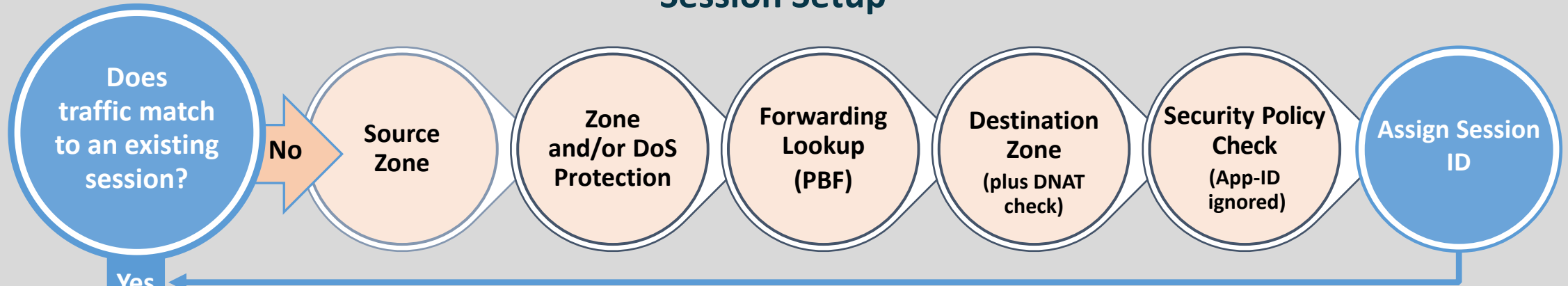
Network address translation

Source NAT configuration

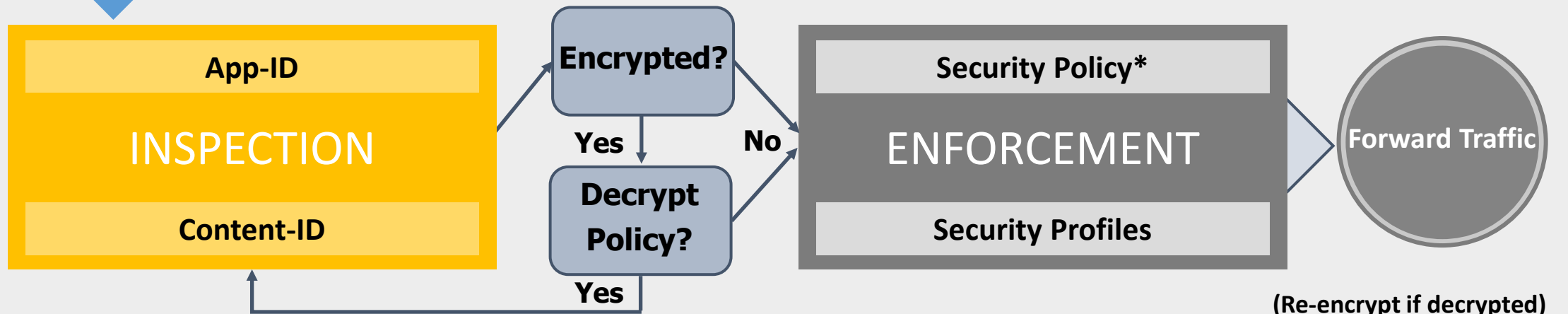
Destination NAT configuration

Flow Logic of the Next-Generation Firewall

Session Setup



Inspection and Enforcement

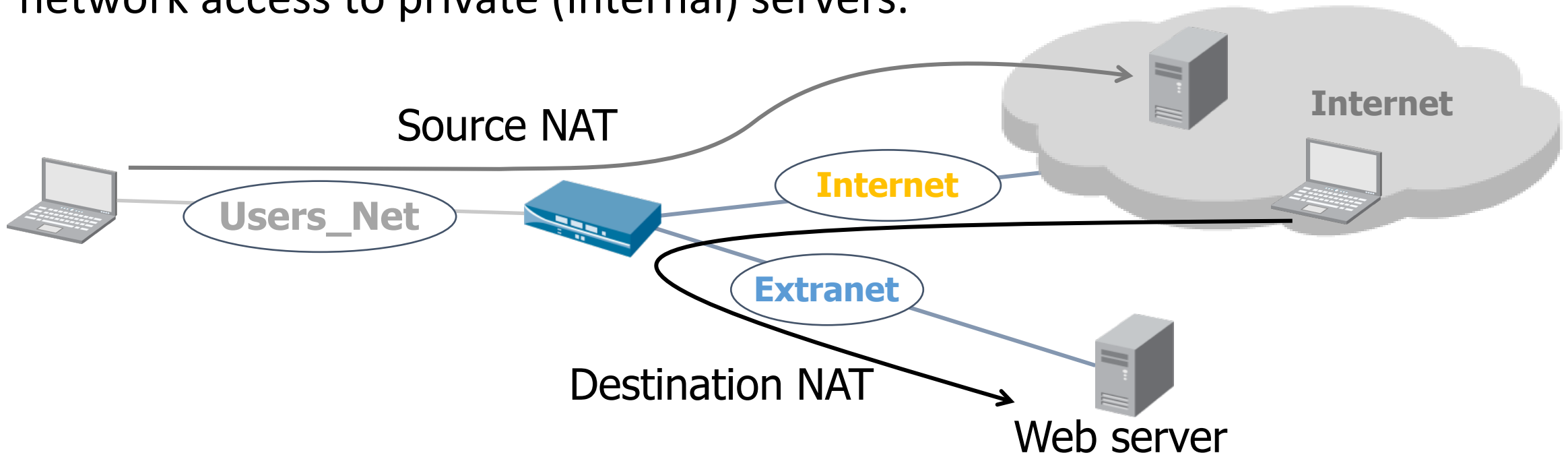


*Policy check relies on pre-NAT IP addresses



NAT Types

- Source NAT commonly is used for private (internal) users to access the public internet (outbound traffic).
- Destination NAT often is used to provide hosts on the public (external) network access to private (internal) servers.



For a Tech Doc about this topic, log into Live and search for "NAT Policy Overview."

Network address translation

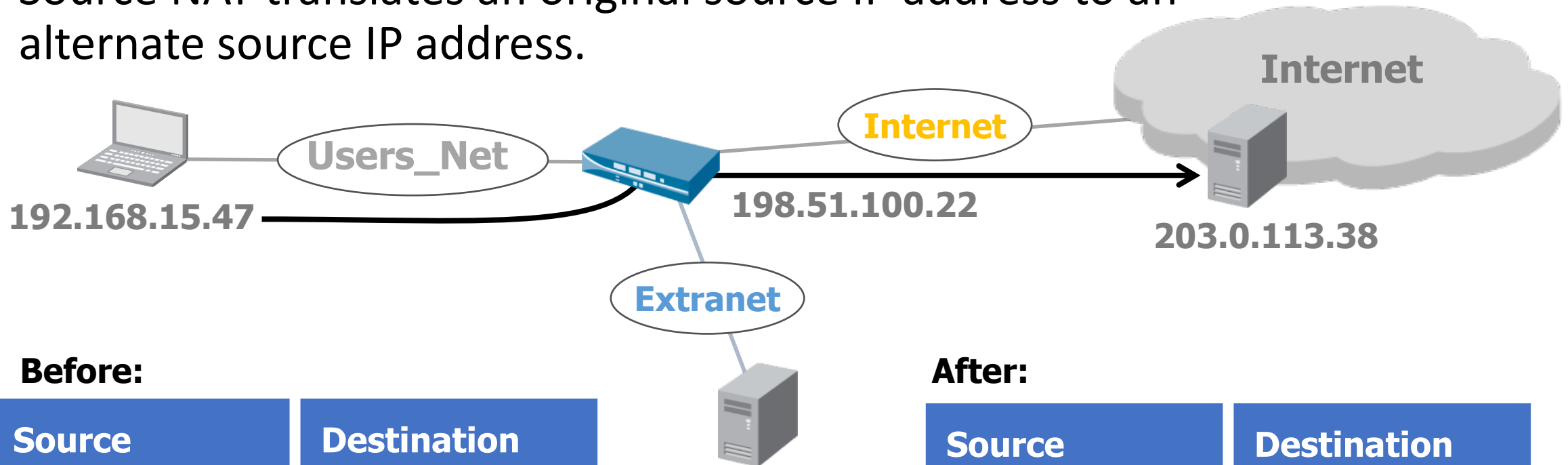


Source NAT configuration

Destination NAT configuration

Source NAT

- Source NAT translates an original source IP address to an alternate source IP address.



Before:

Source	Destination
192.168.15.47	203.0.113.38
Users_Net	Internet

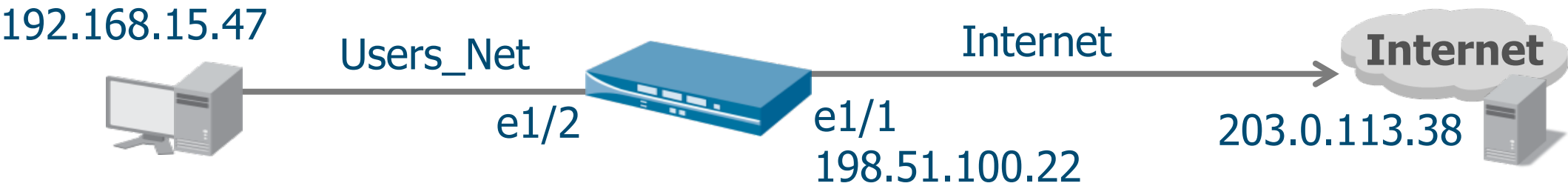
After:

Source	Destination
198.51.100.22	203.0.113.38
Internet	Internet

Source NAT Types

- Static IP:
 - 1-to-1 fixed translations.
 - Changes the source IP address while leaving the source port unchanged.
 - Supports the implicit bidirectional rule feature.
- Dynamic IP:
 - 1-to-1 translations of a source IP address only (no port number).
 - Private source address translates to the next available address in the range.
- Dynamic IP and port (DIPP):
 - Allows multiple clients to use the same public IP addresses with different source port numbers.
 - The assigned address can be set to the interface address or to a translated address.

Source NAT and Security Policies



Policies > NAT

	NAME	TAGS	Original Packet					Translated Packet		HIT COUNT
			SOURCE ZONE	DESTINATION ZONE	DESTINATION INTERFACE	SOURCE ADDRESS	DESTINATION ADDRESS	SERVICE	SOURCE TRANSLATION	
1	Source_Users_Net_to_Internet	Internet	Users_Net	Internet	any	192.168.15.0/24	any	any	dynamic-ip-and-port ethernet1/1 198.51.100.22	283983

Pre-NAT zones

Pre-NAT addresses

Policies > Security

	NAME	TAGS	TYPE	Source		Destination		APPLICATION	SERVICE	ACTION
				ZONE	ADDRESS	ZONE	ADDRESS			
1	Users_to_Internet	Internet	universal	Users_Net	192.168.15.0/24	Internet	any	ssl web-browsing	application-default	Allow

Pre-NAT address

Post-NAT zone

Pre-NAT address

Configure Source NAT

NAT Policy Rule

General | **Original Packet** | Translated Packet

☐ Any	Destination Zone	☐ Any	☒ Any
☒ SOURCE ZONE ^	Internet	☒ SOURCE ADDRESS ^	☐ DESTINATION ADDRESS ^
☐ Users_Net		☐ 192.168.15.0/24	
	Destination Interface		
	any		

+ Add - Delete

Match Criteria

Translation

NAT Policy Rule

General | Original Packet | **Translated Packet**

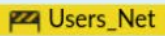
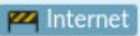
Source Address Translation		Destination Address Translation	
Translation Type	Dynamic IP And Port	Translation Type	None
Address Type	Interface Address		
Interface	ethernet1/1		
IP Address	198.51.100.22/24		

- Dynamic IP And Port
- Dynamic IP
- Static IP
- None

Source NAT Examples

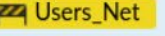
Static 1:1 Translation

Policies > NAT

	NAME	TAGS	Original Packet						Translated Packet	
			SOURCE ZONE	DESTINATION ZONE	DESTINATION INTERFACE	SOURCE ADDRESS	DESTINATION ADDRESS	SERVICE	SOURCE TRANSLATION	DESTINATION TRANSLATION
1	Source_Users_Net_to_Internet	Internet	 Users_Net	 Internet	any	 192.168.15.47	any	any	static-ip 198.51.100.22 bi-directional: yes	none

Dynamic IP Translation

Policies > NAT

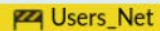
	NAME	TAGS	Original Packet						Translated Packet	
			SOURCE ZONE	DESTINATION ZONE	DESTINATION INTERFACE	SOURCE ADDRESS	DESTINATION ADDRESS	SERVICE	SOURCE TRANSLATION	DESTINATION TRANSLATION
1	Source_Users_Net_to_Internet	Internet	 Users_Net	 Internet	any	 192.168.15.2-192.168.15.50	any	any	dynamic-ip 198.51.100.102-198.51.100.150	none

For a Tech Doc about this topic, log into Live and search for "Getting Started: Network Address Translation (NAT)"

Source NAT Examples (Cont.)

Dynamic IP and Port Translation

Policies > NAT

	NAME	TAGS	Original Packet						Translated Packet	
			SOURCE ZONE	DESTINATION ZONE	DESTINATION INTERFACE	SOURCE ADDRESS	DESTINA... ADDRESS	SERVICE	SOURCE TRANSLATION	DESTINATION TRANSLATION
1	Source_Users_Net_to_Internet	Internet	 Users_Net	 Internet	any	 192.168.15.0/24	any	any	dynamic-ip-and-port 198.51.100.22-198.51.100.23	none

DIPP NAT Oversubscription

- The same translated IP address and port pair can be used multiple times in concurrent sessions:
 - Assumes that hosts are connecting to different destinations

Device > Setup > Session > Session Settings

Session Settings

☒ Rematch all sessions on config policy change

ICMPv6 Token Bucket Size: 100

ICMPv6 Error Packet Rate (per sec): 100

☒ Enable IPv6 Firewalling

☐ Enable Jumbo Frame

☐ Enable DHCP Broadcast Session

NAT64 IPv6 Minimum Network MTU: 1280

NAT Oversubscription Rate: Platform Default

ICMP Unreachable Packet Rate (per sec): Platform Default

☒ Accelerated Aging

Accelerated Aging Threshold: 2x

Accelerated Aging Scaling Factor: 4x

☒ Packet Buffer Protection

Internal Source Port	Firewall Source Port	Destination Address
26435	198.51.100.22:25661	51.6.33.12
35435	198.51.100.22:25661	161.8.55.4
21569	198.51.100.22:25661	201.55.45.1
51043	198.51.100.22:25661	17.39.25.6

Concurrent sessions = oversubscription rate (8/4/2) x address pool size

Network address translation

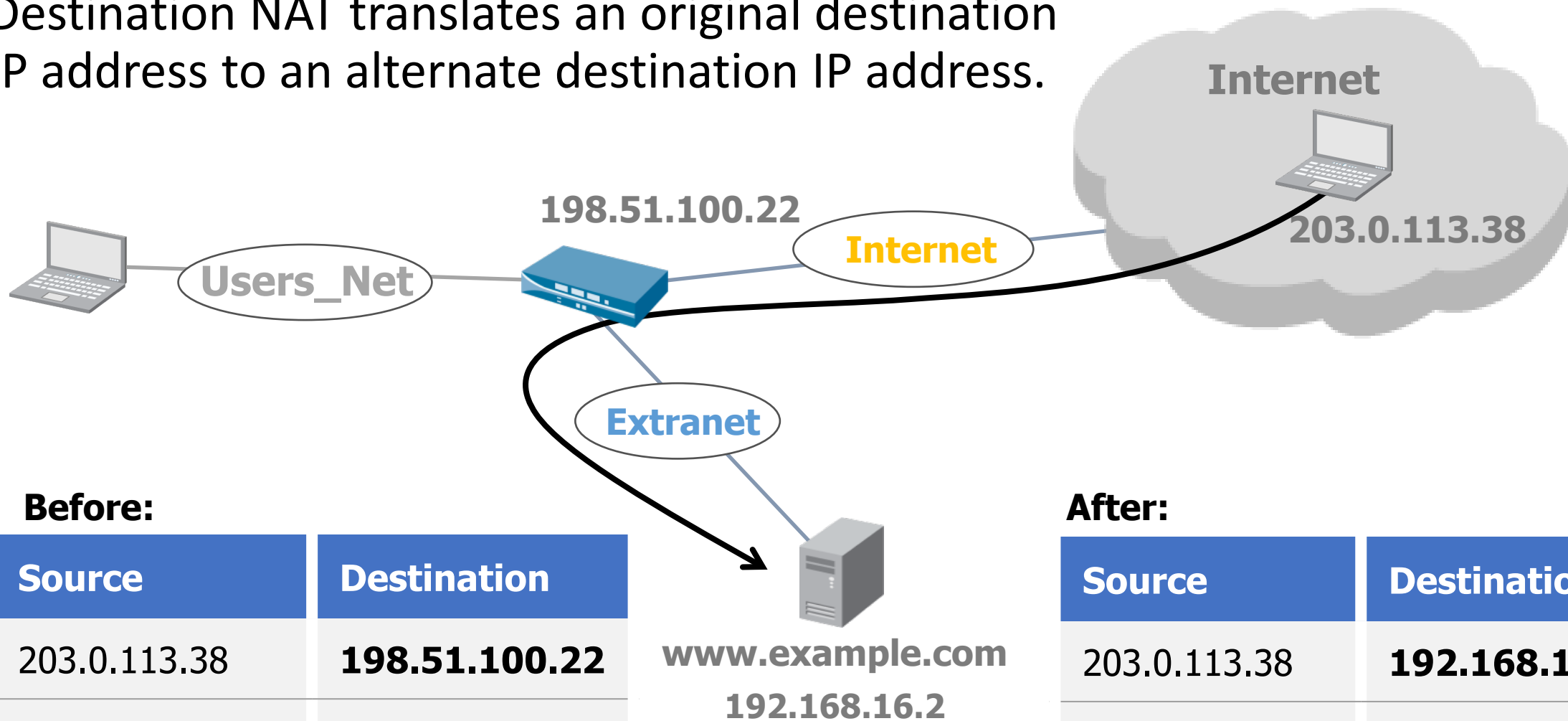
Source NAT configuration

Destination NAT configuration



Destination NAT

Destination NAT translates an original destination IP address to an alternate destination IP address.



Configure Destination NAT

NAT Policy Rule

General | **Original Packet** | Translated Packet

☐ Any	Destination Zone	☒ Any	☐ Any
☒ SOURCE ZONE ^	Internet	☐ SOURCE ADDRESS ^	☐ DESTINATION ADDRESS ^
☐ Internet			☐ 198.51.100.22
	Destination Interface		
	any		

+ Add - Delete

Match Criteria

NAT Policy Rule

General | Original Packet | **Translated Packet**

Source Address Translation	Destination Address Translation
Translation Type: None	Translation Type: Static IP
	Translated Address: 192.168.16.2
	Translated Port: [1 - 65535]
	☐ Enable DNS Rewrite
	Direction: reverse

Translation

Destination NAT Attributes

- Static IP:
 - 1-to-1 fixed translations
 - Changes the destination IP address while leaving the destination port unchanged
 - Also enabled by static source NAT with the **Bi-directional** option set
- Policies > NAT > Add**

NAT Policy Rule

General

Original Packet

Translated Packet

Source Address Translation

Translation TypeNone

Destination Address Translation

Translation TypeStatic IP

Translated Address192.168.16.2

Translated Port[1 - 65535]

☐ Enable DNS Rewrite

Directionreverse

Destination NAT Port Translation Configuration

Policies > NAT

NAT Policy Rule

General

Original Packet

Translated Packet

Source Address Translation

Translation TypeNone

Destination Address Translation

Translation TypeStatic IP

Translated Address192.168.16.2

Translated Port8000

Directionreverse

Used when the destination server is "listening" on a port other than the "well-known" port

	NAME	TAGS	Original Packet						Translated Packet	
			SOURCE ZONE	DESTINATION ZONE	DESTINATION INTERFACE	SOURCE ADDRESS	DESTINATION ADDRESS	SERVICE	SOURCE TRANSLATION	DESTINATION TRANSLATION
1	Dest_NAT_From_Internet	Internet	Internet	Internet	any	any	198.51.100.22	service-http	none	destination-translation address: 192.168.16.2 port: 8000

The service-http option includes TCP port 80 and TCP port 8080.

Dynamic IP Address Support for Destination NAT

- Translates original IP address to destination host with a DHCP-assigned IP address.
- Translated address can be an FQDN, address object, or address group.

Policies > NAT > Add

NAT Policy Rule

General | Original Packet | **Translated Packet**

Source Address Translation

Translation Type: None

Destination Address Translation

Translation Type: Dynamic IP (with session distribution)

Translated Address: PANW-Web-Server

Translated Port: [1 - 65535]

Session Distribution Method: Round Robin

Set translation type to **Dynamic IP**.

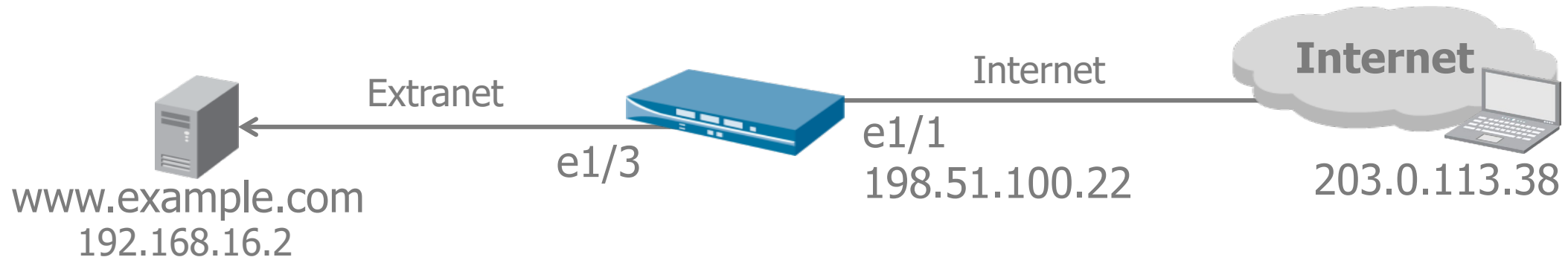
Address

PANW-Web-Server

New Address Address Group

For a Tech Doc about this topic, log into Live and search for "Configure Destination NAT Using Dynamic IP Addresses."

Destination NAT and Security Policies



Policies > NAT

	NAME	TAGS	Original Packet					Translated Packet		
			SOURCE ZONE	DESTINATION ZONE	DESTINATION INTERFACE	SOURCE ADDRESS	DESTINATION ADDRESS	SERVICE	SOURCE TRANSLATION	DESTINATION TRANSLATION
1	Dest_NAT_From_Internet	Internet	 Internet	 Internet	any	any	 198.51.100.22	 service-http	none	destination-translation address: 192.168.16.2

Pre-NAT zones

Pre-NAT address

Policies > Security

	NAME	TAGS	TYPE	Source	Destination		APPLICATION	SERVICE	ACTION
				ZONE	ADDRESS	ZONE			
1	Web_Server_Access	Internet	universal	 Internet	any	 Extranet	 198.51.100.22	 web-browsing	 application-default  Allow

Pre-NAT addresses

Post-NAT zone

Pre-NAT addresses



Configure Bidirectional Source NAT

- Enables internal servers to send and receive traffic through the firewall
- Available only for static NAT

Policies > NAT

NAT Policy Rule

General | Original Packet | Translated Packet

Source Address Translation

Translation Type

Static IP

Translated Address

198.51.100.22

☒ Bi-directional

Destination Address Translation

Translation Type

None

For a Tech Doc about this topic, log into Live and search for "Enable Bi-Directional Address Translation."

Module Summary

Now that you have completed this module, you should be able to:

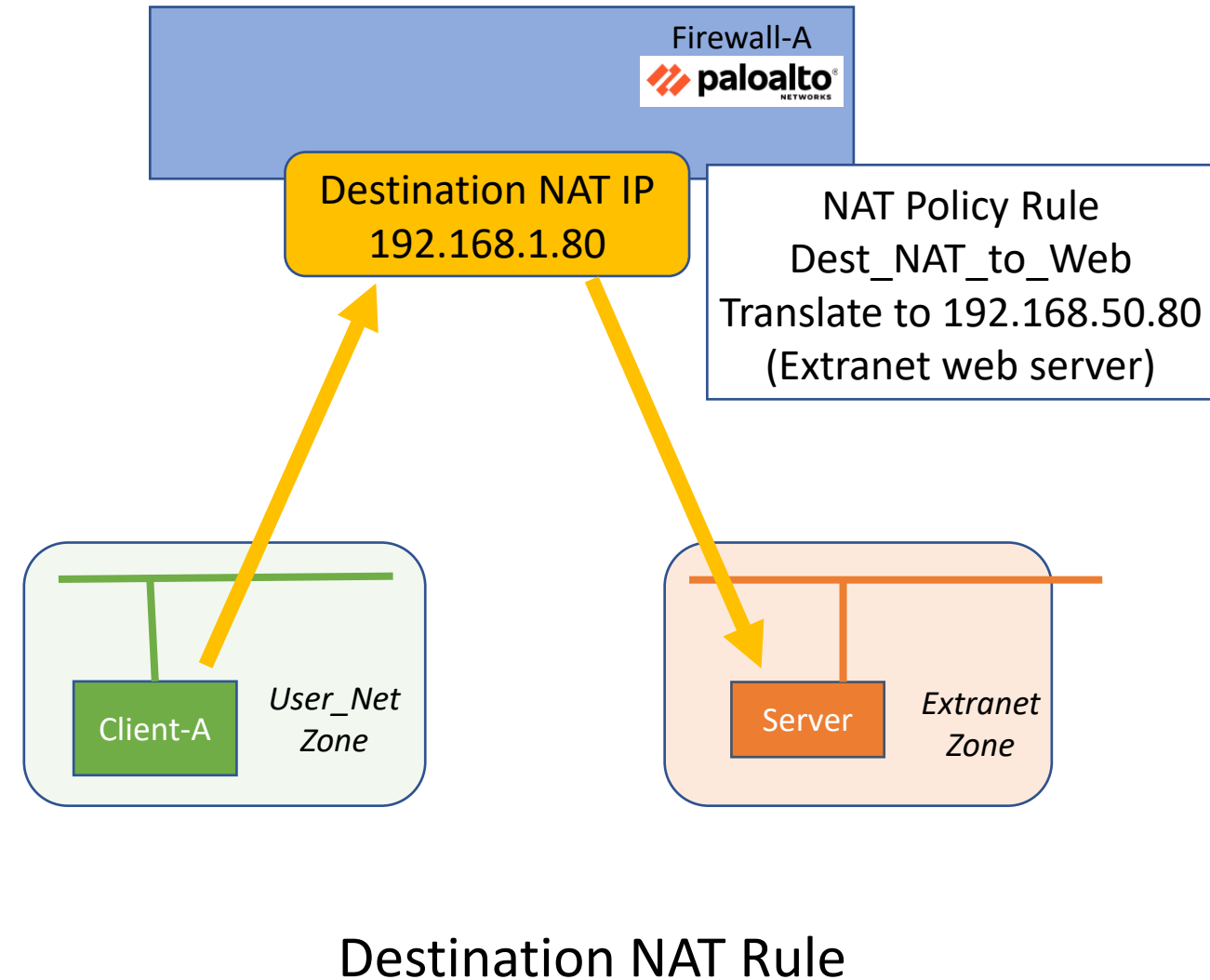
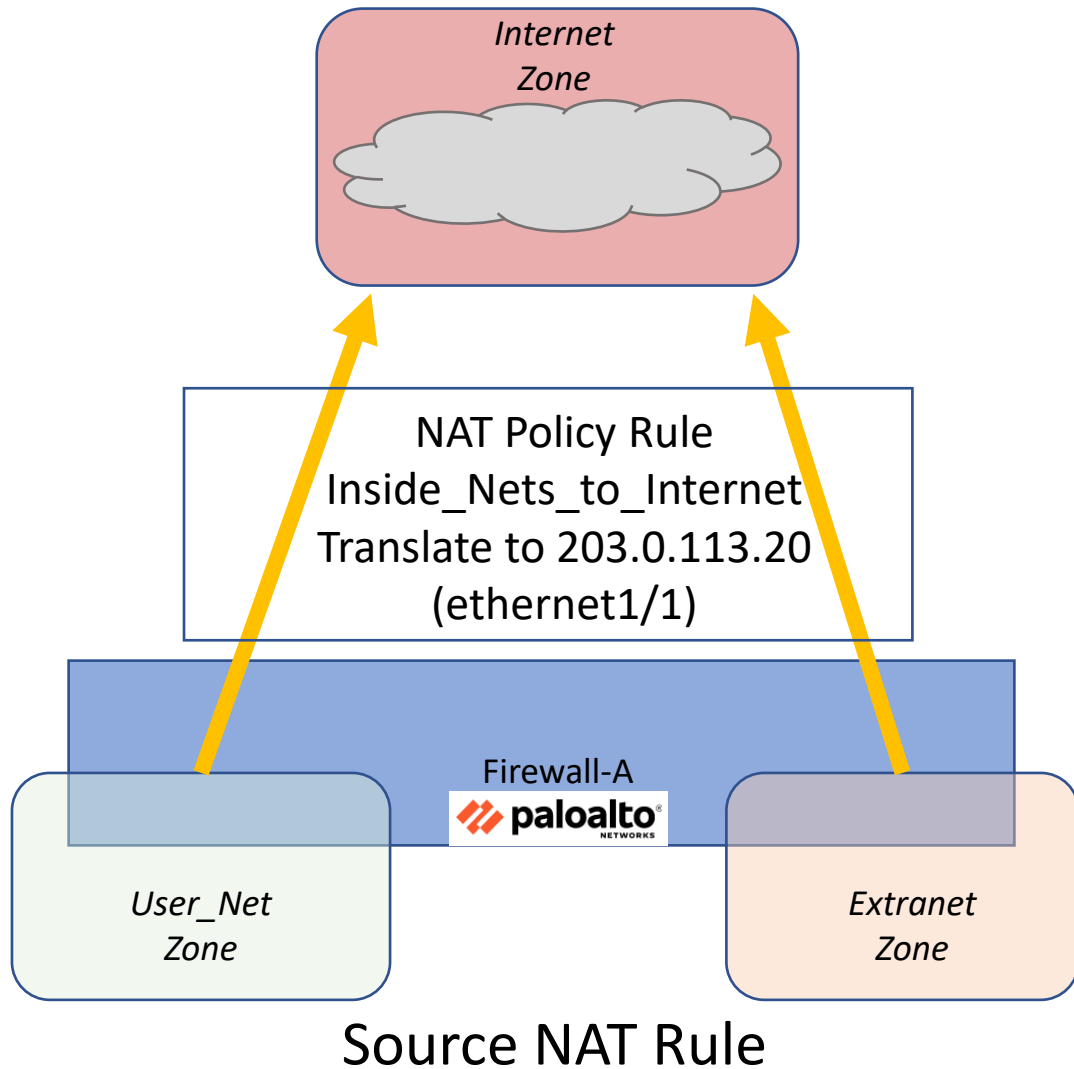
- Configure a NAT policy to implement source NAT
- Configure a NAT policy to implement destination NAT



Questions



Lab 6: Overview



Lab 6: Creating and Managing NAT Policy Rules

- Configure source NAT
- Configure destination NAT

End Content

Unit 7 Homework

- Always check Blackboard for assignments and when they are due.
- Unit 7 Lab: Netlabs Lab 6 – Creating and managing NAT policy rules
- Unit 7 Quiz